Quantification: 536-63x_Hyp_RD_z03c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	21
interior lacuna count	17
frame-edge lacuna count	4

Interior lacunae only: mean, median and sample SD (n = 17)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2443.76	2475	1284.72	px^2
major_axis_length_px	lacuna shape	97.8077	92.9975	49.1755	px
minor_axis_length_px	lacuna shape	38.0502	38.3023	10.1514	px
aspect_ratio	lacuna shape	2.5605	2.415	1.2256	unitless
eccentricity	lacuna shape	0.8665	0.9102	0.102	unitless
solidity	lacuna shape	0.7612	0.7758	0.1264	unitless
roots_count	headline	10.8824	9	7.5984	count
ring_length_r30_px	headline	298.235	273	203.08	px
ring_length_r60_px	reported	589.177	518	402.008	px
owned_length_px	ownership-dependent	662.655	472.676	671.992	px
edge_count	ownership-dependent	34.0588	22	33.9751	count
mean_edge_length_px	ownership-dependent	20.0897	20.4835	7.328	px
perimeter_px	normalised	319.007	298.433	174.187	px
ring_area_r30_px2	normalised	9388.12	8327	4194.2	px^2
ring_area_r60_px2	normalised	21393.2	20339	9675.47	px^2
in_frame_fraction_r30	normalised	0.9781	1	0.0682	unitless
in_frame_fraction_r60	normalised	0.9637	1	0.101	unitless
ring_density_r30	normalised	0.0321615	0.0306579	0.0157715	px^-1
ring_density_r60	normalised	0.0285845	0.0266975	0.0149123	px^-1
roots_per_100px_perimeter	normalised	3.4616	3.4163	1.6357	per 100 px
ring_attached_length_r30_px	network v2	265.294	215	201.292	px
ring_attached_length_r60_px	network v2	436.353	338	375.913	px
ring_length_w_r30_px	network v2	337.417	315.907	233.036	px
ring_length_w_r60_px	network v2	665.9	587.943	459.568	px
sholl_crossings_r10	network v2	7.5882	6	5.455	count
sholl_crossings_r20	network v2	7.4118	7	4.6645	count
sholl_crossings_r30	network v2	7.4706	7	5.6912	count
loop_count	loops, report only	1.1765	0	1.9441	count
loop_edge_share	loops, report only	0.0791	0	0.1085	share
ring_width_mean_r30_px	width	7.2909	7.2343	0.7006	px

Per field
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measure	value	units
analysed_area_px2	982324	px^2
total_skeleton_length_px	21684	px
canalicular_length_density_per_px	0.0220742	px^-1
junction_count	404	count
junction_density_per_px2	0.00041127	px^-2
lacuna_count	21	count

lacunae_per_px2	2.138e-05	px^-2
skeleton_component_count	390	count
mean_component_length_px	55.6	px
median_component_length_px	20	px
field_length_density_w_per_px	0.0249472	px^-1
field_density_without_flagged_per_px	0.022452	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	940	px
unattached_component_count	51	count
canalicular_length_density_attached_per_px	0.0220742	px^-1
field_length_density_with_unattached_per_px	0.0230311	px^-1
loop_count	35	count
loop_edge_share	0.1079	share
edge_count	1186	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.5313	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.7703	px
width_pixels_used	21411	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	1311	70.1057	29.6291	2.3661	0.9063	0.7627	1.4823	11.1503	890.967	True
2	3585	211.287	33.1642	6.3709	0.9876	0.6063	-0.3562	190.729	656.387	False
3	4424	168.508	61.7794	2.7276	0.9304	0.5066	-0.0985	178.037	485.058	False
4	3341	92.9975	56.4772	1.6466	0.7945	0.7441	-0.7147	129.753	115.198	False
5	740	35.5513	27.759	1.2807	0.6248	0.9091	-0.3246	161.099	832.037	False
6	1683	63.5962	39.0607	1.6281	0.7892	0.7571	-0.4294	213.583	88.5187	False
7	2785	110.226	41.3878	2.6632	0.9268	0.6921	-0.2074	250.038	115.956	False
8	1189	59.3046	41.9925	1.4123	0.7061	0.5118	0.1219	253.995	38.2666	False
9	3218	123.087	35.0225	3.5145	0.9587	0.8067	-0.0876	285.404	998.443	False
10	4546	152.562	46.2815	3.2964	0.9529	0.6884	-0.3441	401.046	256.605	False
11	3934	133.278	41.5402	3.2084	0.9502	0.7505	-0.3234	452.159	836.983	False
12	2906	117.901	38.3023	3.0782	0.9458	0.7758	-0.3279	473.994	376.288	False
13	2475	101.315	32.9688	3.0731	0.9456	0.8624	-0.3801	520.411	501.561	False
14	5784	165.174	57.0067	2.8975	0.9386	0.8298	-0.2128	680.673	1000.68	True
15	8146	159.748	75.4405	2.1175	0.8815	0.8502	0.1567	804.35	989.374	True
16	2189	73.1101	40.4728	1.8064	0.8328	0.9132	-0.5214	637.506	936.232	False
17	1176	57.7542	28.7935	2.0058	0.8669	0.8276	-0.2231	683.004	402.083	False
18	679	37.9692	24.2376	1.5665	0.7697	0.8573	-0.1349	734.916	174.795	False
19	829	46.9885	25.6067	1.835	0.8385	0.8399	0.1493	873.288	867.089	False
20	9467	148.982	117.161	1.2716	0.6177	0.7637	0.4415	974.316	939.645	True
21	1845	77.2965	32.007	2.415	0.9102	0.8909	0.0147	929.902	369.506	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	7	241	470	584.032	24	24.3346	7.4473
2	25	436	1059	1229.76	69	17.8226	7.8048
3	31	909	1712	2780.85	143	19.4465	7.2606
4	11	438	920	1280.16	62	20.6477	7.9061
5	7	189	377	181.983	12	15.1652	8.8876
6	9	142	156	218.309	21	10.3957	6.7127
7	8	303	597	856.382	41	20.8874	6.9109
8	15	405	487	381.421	29	13.1525	6.497
9	7	184	421	472.676	20	23.6338	7.1509
10	15	432	905	1159.68	53	21.8807	7.3319
11	13	383	716	707.21	25	28.2884	7.0866
12	11	313	599	309.492	15	20.6328	7.9034
13	9	212	518	348.22	17	20.4835	7.7662
14	2	136	167	86.6985	5	17.3397	
15	4	80	113	50.4558	4	12.614	7.3492
16	2	60	67	28.8995	3	9.6332	
17	9	273	543	479.86	31	15.4794	7.208
18	8	202	508	486.475	22	22.1125	7.5785
19	1	44	71	41.7279	1	41.7279	5.8562
20	0	28	53	0	0	0	4.3619
21	4	145	360	302.037	15	20.1358	6.7922

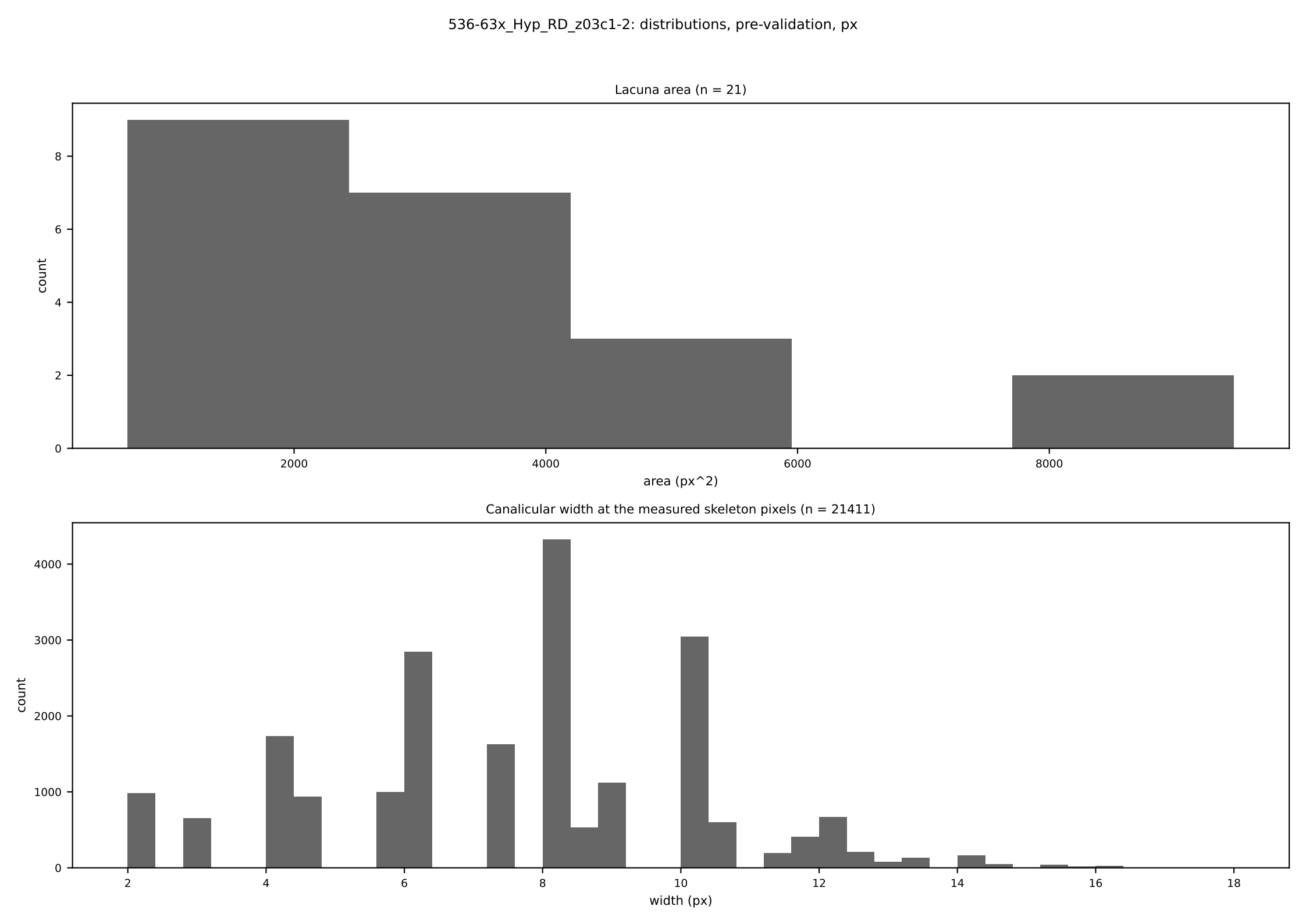
Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	201.782	4986	12541	0.6061	0.574	0.0483353	0.0374771	3.4691
2	655.855	17276	38843	1	1	0.0252373	0.0272636	3.8118
3	733.068	18106	38869	1	1	0.0502044	0.0440454	4.2288
4	398.031	9904	21312	1	1	0.0442246	0.0431682	2.7636
5	114.811	5937	17499	1	1	0.0318343	0.0215441	6.097
6	225.593	2456	2578	1	1	0.0578176	0.060512	3.9895
7	361.789	8165	17200	1	1	0.0371096	0.0347093	2.2112
8	298.433	6199	9201	0.8958	0.756	0.0653331	0.052929	5.0262
9	328.049	8327	18178	0.7322	0.645	0.0220968	0.0231599	2.1338
10	472.659	14091	31679	1	1	0.0306579	0.0285678	3.1735
11	380.534	12648	30304	1	1	0.0302815	0.0236272	3.4163
12	324.635	11576	25943	1	1	0.0270387	0.0230891	3.3884
13	246.551	9801	24158	1	1	0.0216304	0.0214422	3.6504
14	460.794	5167	7599	0.5692	0.5578	0.0263209	0.0219766	0.434
15	525.623	8185	11797	0.6487	0.6114	0.00977398	0.00957871	0.761
16	224.693	7360	14921	1	0.995	0.00815217	0.00449032	0.8901
17	188.93	7444	20339	1	1	0.0366738	0.0266975	4.7637
18	120.125	5980	17550	1	1	0.0337793	0.0289459	6.6597
19	144.368	5916	13005	1	1	0.00743746	0.00545944	0.6927
20	703.76	10275	13592	0.6197	0.5585	0.00272506	0.00389935	0
21	204.995	8412	22105	1	0.9866	0.0172373	0.0162859	1.9513

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	183	432	272.894	529.304	3	8	5
2	366	870	490.517	1198.23	15	10	11
3	887	1578	1041.62	1954.8	21	19	24
4	411	766	490.333	1043.41	12	13	13
5	120	178	204.225	411.764	3	6	4

6	141	141	158.882	174.539	3	3	0
7	282	470	341.664	677.5	9	9	8
8	366	430	459.848	547.789	10	5	5
9	180	291	201.368	466.291	5	4	5
10	406	624	485.304	1014.58	13	12	12
11	358	543	435.889	806.796	10	9	8
12	215	272	364.676	685.269	5	10	9
13	210	338	238.823	587.943	6	7	7
14	108	108	145.74	176.225	3	5	1
15	47	47	85.9411	127.225	4	0	2
16	32	32	63.0416	70.8701	1	1	0
17	233	371	315.907	635.441	8	8	7
18	123	247	233.894	572.831	3	6	8
19	40	49	47.799	76.1127	1	1	2
20	0	0	27.3137	54.4558	0	1	4
21	140	218	162.297	396.149	4	3	4


536-63x_Hyp_RD_z03c1-2: distributions, pre-validation, px
Lacuna area (n = 21)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 21411)
count
width (px)